

Sebastian Vargas

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EDUCATION

University of California, Berkeley - College of Engineering

Berkeley, CA

B.S. Electrical Engineering & Computer Science

Expected Graduation: May 2028

- **GPA:** 4.00/4.00 | **Units:** 80.44 / 120 | **Awards:** [Dean's List](#), [Honors to Date](#), Eta Kappa Nu ([HKN](#), EECS)
- **Coursework:** Deep Learning, Computer Vision, Probabilistic and Learning-Based Navigation for Physical AI, Robotics, Algorithms, Operating Systems, Modeling Networks, Data Structures, Computer Architecture, Discrete Math & Probability, Signals & Systems

SKILLS

Languages: Python, Java/Kotlin (backend), JavaScript/TypeScript/HTML/CSS (frontend), C/Rust (systems), SQL (databases), RISC-V

ML & Robotics: ROS 2, scikit-learn, NumPy/Pandas, LeRobot, HuggingFace, MoveIt, PX4, Linux, PyTorch, OpenCV

Systems & Cloud: AWS (Bedrock, Knowledge Bases), Azure, PostgreSQL, MongoDB, Firebase, REST APIs, React/Node.js

PROFESSIONAL EXPERIENCE

Google | Software Engineering Intern | San Jose, California

May 2026 - Aug 2026

- Developed novel 2FA method for Google Sign-In, removed barriers for users to increase successful sign-in rate using SMS systems.
- Owned project scope end-to-end: research and design phase, backend integrations, updated user experience, and launch process.
- Collaborated with cross-departmental stakeholders to build out design and integrate into existing 2FA systems with full test suite.

Google | Associate Software Developer Intern | New York, New York

May 2025 - Aug 2025

- Developed change history functionality for Google's internal performance management system, processing 10,000+ quarterly performance records and enabling users to view modifications to quarterly expectations and tasks with full audit trail capabilities.
- Owned the full lifecycle with end-to-end implementation, including building APIs to pull change history from Google internal datastores and developing reusable front-end components reducing development time by 30%.
- Designed and deployed role-based access control (RBAC) system with OAuth 2.0 authentication for 150,000+ users.

NextEra Energy | AI Technical Project Manager, Contract | Juno Beach, Florida

Sep 2025 - Jan 2026

- Architected "Renewables RegWatch," an enterprise RAG (Retrieval-Augmented Generation) AI solution leveraging AWS Bedrock and Amazon Knowledge Bases to deliver automated regulatory intelligence, reducing manual research time by 75%.
- Managed cross-functional team of 5 engineers developing scalable web scraping infrastructure (Beautiful Soup, Selenium), ETL data pipeline, vector database integration with AWS Knowledge Bases, automated PDF report generation, and email distribution system.

RESEARCH & EXTRACURRICULARS

UC Berkeley MSC Lab | Undergraduate Researcher | Berkeley, California

Sep 2026 - Present

- Conduct research on physics-based simulation and verification methods for dexterous robotic manipulation systems.
- Evaluate and extend UniLab for sim-to-real workflows and support across diverse robotic manipulator platforms.

Residual RL Post-Training on Vision-Language-Action Models | [Personal Research Project](#)

Summer 2026

- Fine-tuned a 450M-param VLA (SmolVLA) on self-collected data, designed a residual RL system (SAC) on top of the frozen policy.
- Built a staged 4-class reward classifier (dual ResNet-18 + MLP head) to replace binary labeling, improving held-out accuracy 74.3%→82.5% and cutting calibrated NLL 24% via temperature scaling.
- Re-engineered HuggingFace's HIL-SERL infra to support residual RL on a frozen VLA rather than learning from scratch, bridging joint-space VLA outputs and Cartesian-delta RL actions via forward kinematics.

Tiltrotor UAV Control | Advised by Sastry Lab Researcher | [Research Project Website](#)

Mar 2026 - May 2026

- Implemented and tested custom control loops in simulation for overactuated tiltrotor drone verifying full SE(3) capabilities.
- Wrote ROS2 nodes and updated PX4 firmware to enable drone allocation matrix to properly identify physical geometry for flight.
- Performed Hardware-In-The-Loop testing and analyzed sensor readings to validate control stack and proper implementation.

UC Berkeley EECS Department | [EECS C106A](#) Content Assistant

Aug 2026 - Present

- Collaborate with head Content TA to develop curriculum and lead weekly homework parties and grading for UC Berkeley's Introduction to Robotics, a course with 200+ students covering kinematics, dynamics, controls, computer vision, and sensing.
- Created robust coding portion of 10+ assignments to support programming skillsets of students for holistic robotics learning.

UpSync Consulting | Vice President of Projects

Sep 2025 - Present

- Led delivery of 5 client projects valued at \$100,000, managing a team of 12 strategy and technical PMs.
- Engineered automation of sourcing workflows, processing ~10,000 emails weekly with 70% reduction in manual effort.
- Created partnerships with professionals at industry firms, including Oracle, Adobe, Pinterest, Fenway Sports Group, and AWS.
- Consulted for AMD and Lenovo on full-stack development and AI-assisted software workflows, improving productivity by 40%.